

RIZKY DWI SAPUTRA

Hardware Interface Programmer | .NET/C# | Medical & Laboratory Systems | Embedded & IoT

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PROFILE

Hardware Interface Programmer and R&D Engineer with experience spanning software development, medical/laboratory instrument integration, embedded systems, electronics, networking, and on-site deployment. Experienced in building desktop and web-based systems, connecting instruments through Serial RS232/TCP-IP, implementing ASTM/HL7 interfaces, bridging data to LIMS through REST APIs, and supporting production deployments. Also experienced in hardware prototyping, PCB work, microcontrollers, RFID, robotics, and technical troubleshooting.

CORE SKILLS

- **Software:** C#, .NET, Python, Java, Kotlin; desktop and embedded applications.
- **Integration:** Serial RS232, TCP/IP Socket, ASTM, HL7, REST API, LIMS/LIS, hardware interfaces.
- **Data:** PostgreSQL, Redis Server, rabbitmq backup & restore.
- **Hardware:** Microcontroller, ESP/Arduino, RFID, IoT, robotics, PCB design (KiCad/Eagle), electronics service.
- **Delivery:** FAT, SIT, UAT, production deployment, technical reporting, technical documentation, troubleshooting.

PROFESSIONAL EXPERIENCE

Mar 2024–Present | PT PentaIndotek Mandiri, Bandung Lead Hardware Interface Programmer | Full-time

- Develop and maintain laboratory instrument applications and bridge instruments to LIMS.
- Develop and maintain embedded applications, hardware components/circuit boards, and multiple hardware modules and communication protocols.
- Collaborate with backend developers and other programmers to integrate hardware functionality.
- Test hardware components individually or with principals; complete FAT, SIT, and UAT before production deployment.
- Designed, developed, and managed a HealthTech software ecosystem for laboratory management, implemented in 40+ healthcare facilities in Indonesia.
- **ins.care & inscare+:** .NET web-based middleware for centralized laboratory data, TCP/Serial instrument communication, and REST API bridging to LIMS.
- **ei-link:** Python desktop application for low-level hardware interaction on local laboratory computers.
- **PetaIn:** centralized monitoring dashboard for application health, instrument connection status, and middleware status across client sites.
- **Helpio:** internal ticketing/incident-management system for technical issues, bug tracking, and healthcare-site support workflow.
- **Gynocam:** R&D application for early cervical-cancer screening; currently in operational pilot testing at RS Permata Cibubur and a network of clinics.

Dec 2022 – Feb 2024 | PT Nutech Integrasi, Jakarta Hardware Interface Programmer | Full-time

- Develop and maintain desktop and embedded applications and hardware/circuit-board components.
- Handle hardware modules and communication protocols; collaborate on hardware/software integration.
- Perform component testing, FAT/SIT/UAT, production deployment, technical reporting, and technical-data preparation.
- **Immigration Gate System:** magnetic gate controller interface SDK using Serial, Socket, and C# Desktop.
- **Reska Parking System:** reader-device agent using Serial, Socket, C# Desktop, Java, and AES-256.
- **E-Money Card Vending Machine:** C# WPF interface with Serial/Socket, card reader, QR system, PostgreSQL, and card dispenser.

Aug – Nov 2022 | PT Teknologi Komponen Dalam Negeri, Jakarta Intern — Medical Device Project

- Developed prototypes for thermometer, blood-pressure monitor, and weight/height scales using ESP microcontrollers, interface systems, and desktop applications.

Apr 2021 – Jul 2022 | PT Applimetis Parama Solusi, Bekasi
R&D Engineer | Full-time

- Developed embedded applications using ASTM and HL7.
- Designed and developed hardware components/circuit boards and prototypes.
- Handled hardware modules and communication protocols; tested biomedical laboratory devices.
- Deployed LAN/WAN/local network infrastructure at project sites and completed SIT/UAT and production deployment.
- **MedQlab**: interface programming for hospital laboratory devices information systems.
- **MedQV**: camera project for identifying diseases in female organs, including cervical cancer and HIV/AIDS.
- **ACCMS Aerocom Tube Pneumatic System**: RFID hardware assembly and microcontroller programming.

Jan – Jul 2020 | PT Tamano Indonesia, Karawang
Assembly | Full-time

- Assembly and measurement, quality control, equipment maintenance, troubleshooting, robotics machine setup/maintenance, production records, and coordination across assembly stations.

Sep – Dec 2019 | PT Puji Tri Daya, Bandung
Field Coordinator Intern

- Supervised contractor quality, budget, schedule, materials, field team, billing checks, project calculations, site mapping, and coordination with external parties. Projects included SUTM/SUTR electrical infrastructure at TPPAS Legok Nangka and Jatigede Dam.

Jun 2017 – Jul 2018 | CV Netral, Bandung
IT Support Helpdesk | Full-time

- Computer/application support, OS and application updates, backup/restore, printer/scanner support, security checks, hardware/network troubleshooting, and data management/data entry for PLN Region Bandung.

Feb 2014 – May 2017 | Inorobo Robotics School, Bandung
Trainer | Part-time

- Prepared technical training and learning materials; taught programming applications, electronics, Arduino, ESP8266, ATmega, servo, power consumption, Node-RED, problem solving, and project activities.

Sep – Nov 2012 | CV Jeck'sindo Electronics, Jakarta
Intern

- Electronics troubleshooting and repair, power-electronics maintenance, telephone/PABX network repair, service/monitoring, and reporting.

SELECTED SIDE PROJECTS / FREELANCE

- **2023 — Plaza Medika Jakarta**: LIS integration with hematology and urinalysis laboratory equipment using C# desktop service.
- **2023 — Klinik Utama Glori Medika Sunter**: LIS integration with hematology, blood-chemistry, and urinalysis analyzers using C# desktop service.
- **Electronics services**: weekend repair/troubleshooting of phones, electric fans, laptops, and other electronics.

EDUCATION

2017 — D-IV Informatics Management, Politeknik Piksi Ganesha Bandung

2013 — Industrial Electronics Engineering, SMKN 1 Losarang Indramayu

TRAINING & CERTIFICATES

- Memulai Pemrograman dengan Kotlin — Dicoding (Dec 2023)
- Python Data Science — DQLab Academy (Feb 2022)
- AWS Cloud Practitioner Essentials — Dicoding (Apr 2021)
- Dasar Mobile Programming dan Kotlin — Skill Academy (Nov 2020)
- Japanese Language Course — LPK Seiko Indramayu (Mar 2019)
- Cyber Security “Secure Apps” — CyberArmy Bandung (Nov 2018)
- Paper-based TOEFL — SWIFT English (Sep 2018)
- Microsoft Office Specialist — Certiport (Sep 2017)
- Engineering Information System Certificate Competence (Jul 2017)
- Workshop DJI Drone — IndoDrone (Apr 2017)
- Information Technology Network Certificate Competence (Jul 2016)
- Information Technology Support Certificate Competence (Jul 2015)
- Industrial Electronics Engineering Expertise — Electro (Oct 2013)